

agent seeks to maximize its cumulative reward. Let vector $\mathbf{b} = (b(1), b(2), \dots, b(|\mathcal{S}|))$ denote the belief state where $b(s)$ is the probability that the true system state is $s \in \mathcal{S}$. Starting from belief $\mathbf{b}$ at time t , the agent updates belief to $\mathbf{b}'$ at time $t + 1$ after executing action a and observing o as

$$b'(s') = \frac{P(o|a, s')}{P(o|\mathbf{b}, a)} \sum_{s \in \mathcal{S}} P(s'|s, a)b(s) = \frac{P(o|a, s') \sum_{s \in \mathcal{S}} P(s'|s, a)b(s)}{\sum_{s' \in \mathcal{S}} P(o|a, s') \sum_{s \in \mathcal{S}} P(s'|s, a)b(s)}, \forall s' \in \mathcal{S}. \quad (2.1)$$

Naturally, $\sum_{s \in \mathcal{S}} b(s) = 1$. The initial belief, $\mathbf{b}_0$, provides the probability distribution of the state at the beginning of the planning horizon. The beliefs derived from an initial belief, via a feasible sequence of actions and observations are called *reachable belief points*.

Figure 1 shows the belief states that are *reachable* from the initial belief state, $\mathbf{b}_0 = (0.5, 0.5)$ in one stage for the well-known *tiger problem* presented by Kaelbling et al. (1998), considering three actions, Listen (a_1), Open Left Door (a_2), Open Right Door (a_3), and two observations resulting from each action, under the assumption that opening a (left or right) door restarts the problem and resets the belief to $(0.5, 0.5)$. Appendix B contains a detailed description of the tiger problem.

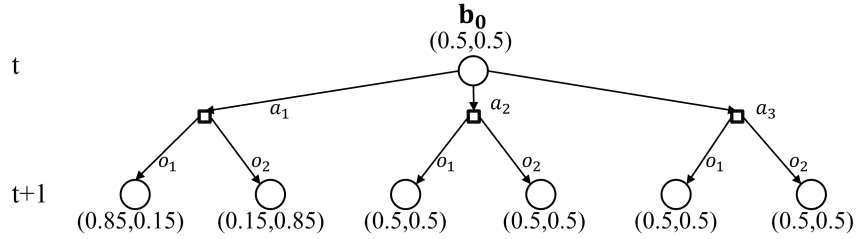


Figure 1: Tree structure of a two-stage tiger problem (Kaelbling et al. 1998). Circles represent belief states observed right before taking an action in each stage.

A policy π provides the sequence of actions to be taken over the planning horizon as a function of the belief state. The value function represents the expected cumulative reward obtained under the optimal policy, π^* . For a finite horizon problem, the following backward recursion computes the value function, $V_t(\mathbf{b})$, as,

$$V_t(\mathbf{b}) = \max_{a \in \mathcal{A}} \left[\sum_{s \in \mathcal{S}} b(s)R(s, a) + \sum_{o \in \mathcal{O}} P(o|\mathbf{b}, a)V_{t+1}^{\pi^*}(\mathbf{b}') \right], \forall \mathbf{b} \in \mathcal{B}_{\sqcup}, \sqcup \in \{t, \infty, \dots, \mathcal{T} - \infty\}, \quad (2.2)$$

where $\mathcal{B}_{\sqcup}$ denotes the $|\mathcal{S}|$ -dimensional belief space. The optimal policy for belief state $\mathbf{b}$ at time period t is defined as

$$\pi_t^*(\mathbf{b}) = \arg \max_{a \in \mathcal{A}} \left[\sum_{s \in \mathcal{S}} b(s)R(s, a) + \sum_{o \in \mathcal{O}} P(o|\mathbf{b}, a)V_{t+1}(\mathbf{b}') \right]. \quad (2.3)$$

Sondik (1971) showed that the value function is represented exactly by a piecewise-linear and convex function such that the value function for a specific period is represented by a set of $|\mathcal{S}|$ -